Do Now · retrieval

Lesson 103 · Input and Output: Selecting Devices for Tasks

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What does an input device do?

Q2

What does an output device do?

Q3

What is an actuator?

Q4

A 2×3 display grid: how many cells?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — An input device captures data or a user action; an output device communicates information or changes physical conditions.

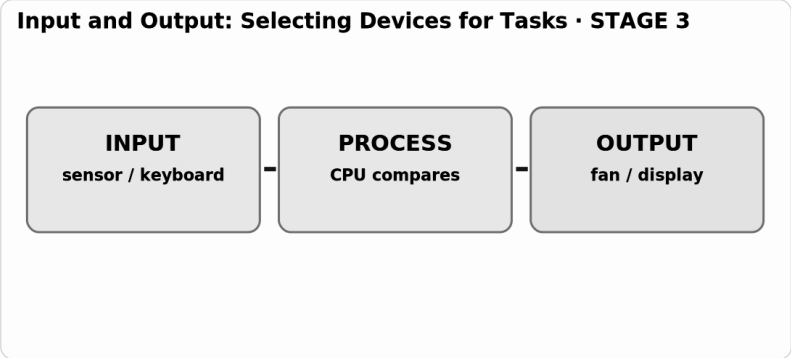
Core idea 2

Apply — A temperature sensor measures physical conditions; an actuator controls a physical action; a display shows values to the user.

Core idea 3

Evaluate — Choose devices by required accuracy, accessibility, environment and response time. A touchscreen can perform both input and output functions.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

input — Data entering computer system.

output — Information/effect produced.

sensor — Measures a physical quantity.

actuator — Creates a physical action.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

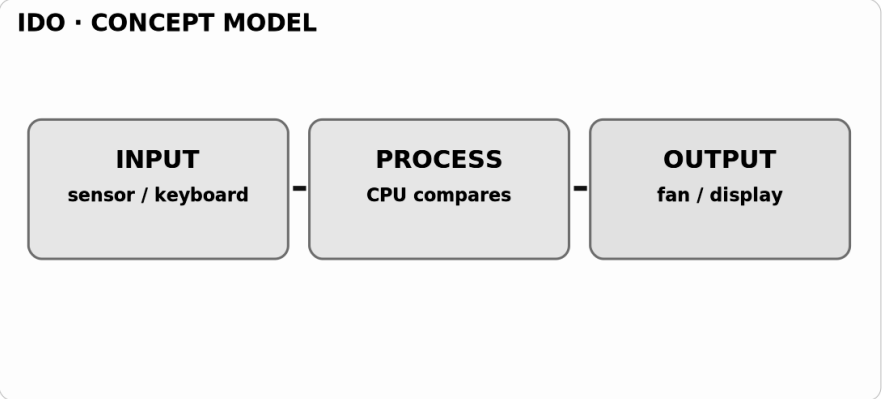
Problem

Recommend input and output for automated greenhouse temperature control.

Follow the worked method

1. Explain the choice — Input thermistor/temperature sensor; output fan/heater actuator, possibly a display.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Explain the choice
Input barcode reader or RFID reader; output display and optional printer.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

For a room-temperature controller, choose appropriate input and output devices and explain roles. [6]

Question 2

Compare barcode reader and RFID for counting library books, linking benefits to context. [6]

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Design accessible touchscreen ticket kiosk: give two input and two output features and justify. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A touchscreen can only be an output device.

Use key vocabulary from memory

1. Explain input in your own words.

2. Explain output in your own words.

3. Explain sensor in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.